

NVIDIA

NASDAQ : NVDA · Mature-Company DCF \$4801.2B mkt cap · 24.30B sh · \$10.6B cash, \$8.5B debt · \$10.6B cash vs ~\$8.5B debt (~\$2B net cash); ~24.3B shares; +\$80B buyback authorized, \$0.25/quarter dividend; Revenue \$215.9B FY26 (+65%); GAAP op margin ~60%; FCF ~\$96.7B (~45%); Data Center now ~88% of revenue; World's largest company at ~\$5.0T; ATH ~\$235 (May), ~\$205 now (~+45% TTM); slid post-earnings despite the beat

\$197.58

THE CENTRAL QUESTION

Can datacenter AI capex compound for years at high-70s% gross margin to justify ~\$5T and ~50x FCF, or does a digestion/air-pocket hit a stock priced for perfection?

NVIDIA trades at ~\$4,986B (~\$205/sh, ~+45% TTM) - ~50x FCF on \$215.9B revenue (+65%), ~60% operating margins, ~\$2B net cash. Data Center is ~88% of revenue and grew +92% last quarter at a mid-70s% gross margin. The DCF asks whether hyperscaler + sovereign AI capex keeps compounding or pauses; even at huge growth the modal can't justify ~\$5T - weighted -25% (ultra-bear -80% to ultra-bull +73%).

PROBABILITY-WEIGHTED EXPECTED VALUE

\$153.54 expected fair value today
-22.3% vs spot \$197.58

FORWARD COMPOUNDED VALUE

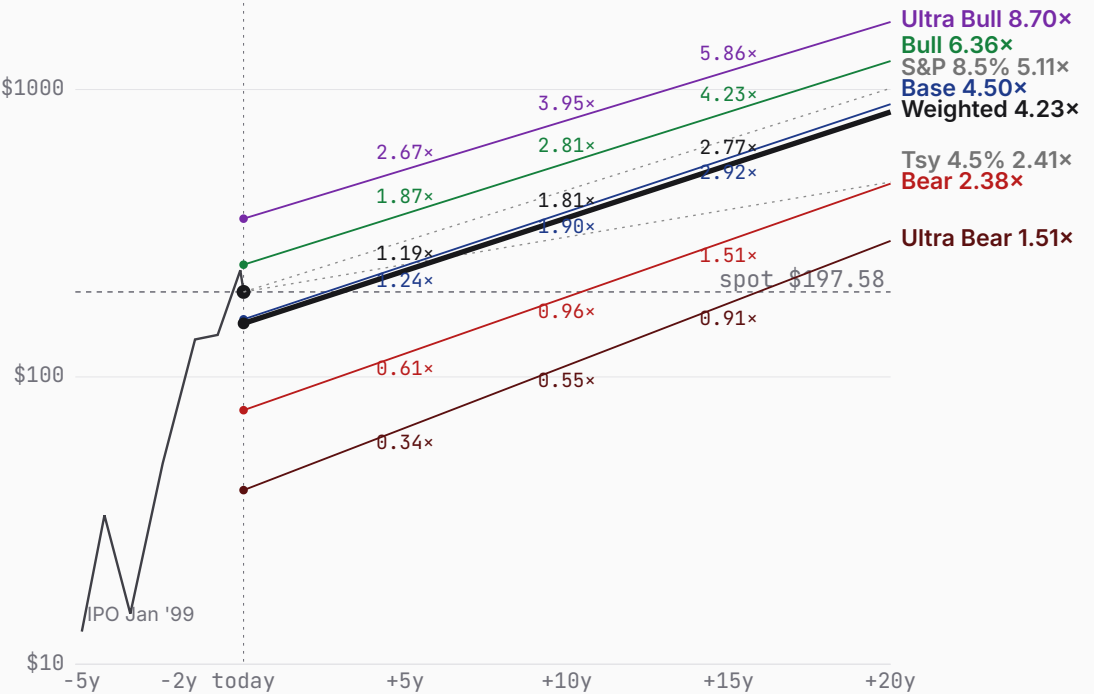
+5y	+10y	+15y	+20y
\$234.28	\$357.69	\$546.47	\$835.42
1.19x spot	1.81x spot	2.77x spot	4.23x spot

WEIGHTING RATIONALE

- Ultra Bear 15% Capex air-pocket; revenue contracts; ~\$40 (-80%).
- Bear 25% Growth decelerates hard; share erodes; ~\$77 (-63%).
- Base 32% Capex holds; cadence extends; still rich; ~\$159 (-23%).
- Bull 20% Capex holds; Rubin extends cadence; ~\$246 (+20%).
- Ultra Bull 8% AI build-out runs for years; ~\$355 (+73%).

Bull (20%) + Ultra Bull (8%) together contribute \$77.57 — 51% of the \$153.54 expected value despite 28% combined probability. Today's spot (\$197.58) sits 29% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$40.37	BEAR	\$76.59	BASE	\$158.65	BULL	\$245.72	ULTRA BULL	\$355.27
	-79.6% vs spot		-61.2% vs spot		-19.7% vs spot		+24.4% vs spot		+79.8% vs spot
CAGR -0.6% WACC 10.5% CAGR -0.6% Prob 15%		CAGR +7.6% WACC 9.5% CAGR +7.6% Prob 25%		CAGR +20.1% WACC 9.0% CAGR +20.1% Prob 32%		CAGR +25.3% WACC 8.5% CAGR +25.3% Prob 20%		CAGR +32.0% WACC 8.2% CAGR +32.0% Prob 8%	
Capex air-pocket; revenue contracts.		Growth decelerates hard; share erodes.		Capex holds; cadence extends; still rich.		Capex holds; Rubin extends the cadence.		AI build-out runs for years; full platform.	

AR
2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

NVIDIA scenario narratives

PROBABILITY WEIGHTS

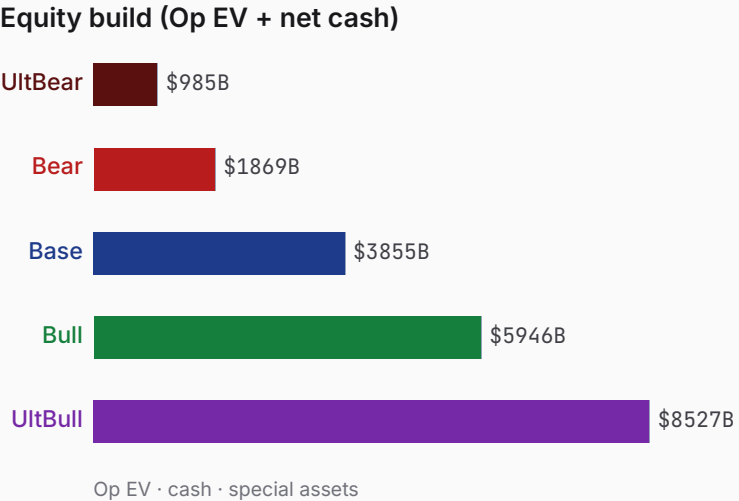
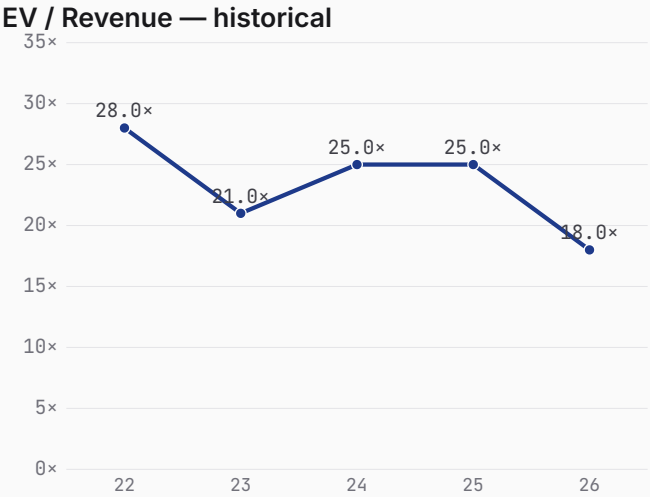
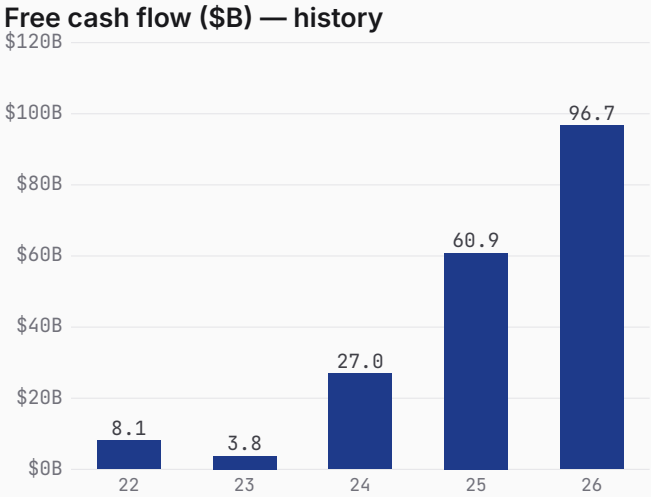
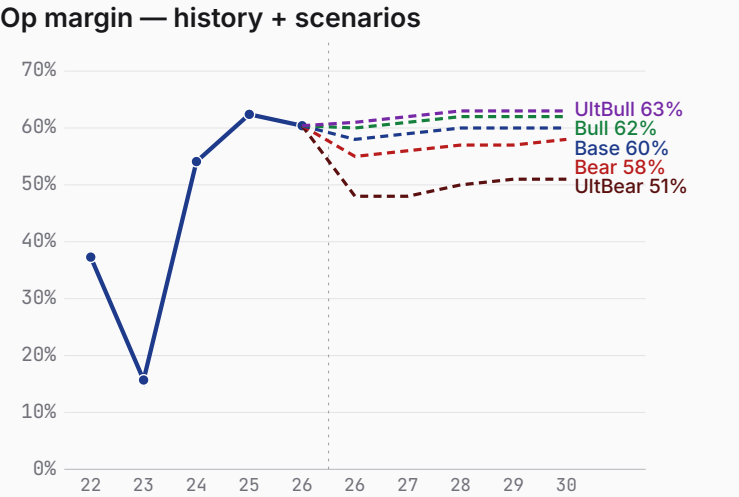
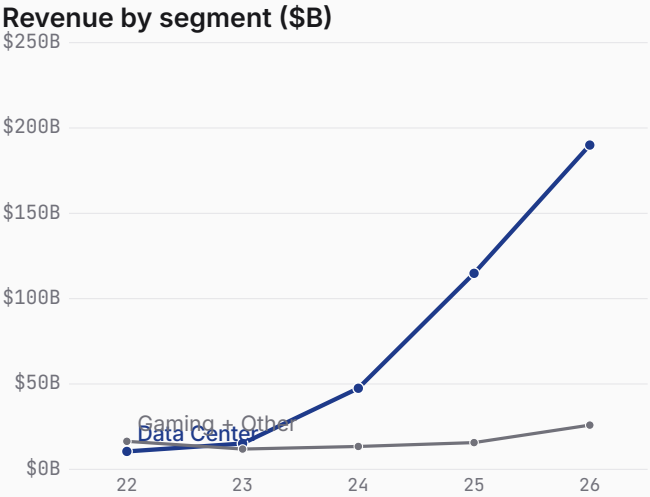
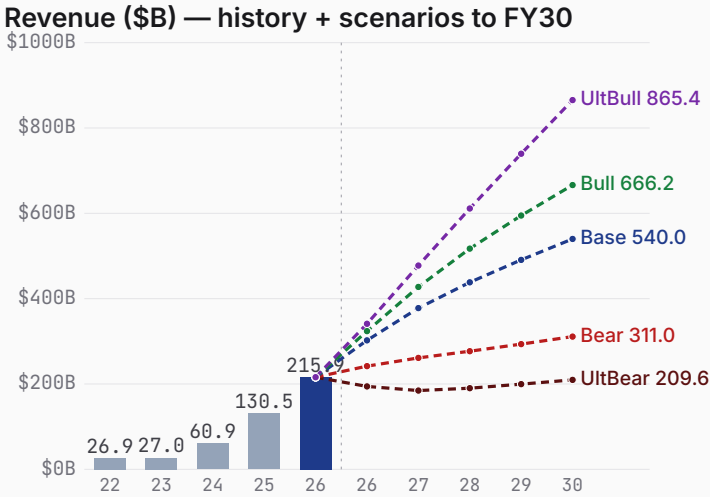
Ultra Bear 15% Bear 25% Base 32% Bull 20% Ultra Bull 8% · Weighted expected \$153.54 (-22.3% vs spot)

ULTRA BEAR	\$40.37	BEAR	\$76.59	BASE	\$158.65	BULL	\$245.72	ULTRA BULL	\$355.27
	-79.6% vs spot		-61.2% vs spot		-19.7% vs spot		+24.4% vs spot		+79.8% vs spot
Probability 15%		Probability 25%		Probability 32%		Probability 20%		Probability 8%	
Capex air-pocket; revenue contracts.		Growth decelerates hard; share erodes.		Capex holds; cadence extends; still rich.		Capex holds; Rubin extends the cadence.		AI build-out runs for years; full platform.	
WHY 15%		WHY 25%		WHY 32%		WHY 20%		WHY 8%	
The genuine bear: the entire thesis rides on hyperscaler capex holding, custom silicon (OpenAI-Broadcom 10GW) + AMD (Meta ~\$100B, OpenAI 6GW) all land H2-2026, and the stock already slid at \$5T despite a +85% quarter. 15% weight on a real digestion/share-loss break.		Growth deceleration plus custom-silicon/AMD share loss is the central pull on a name lapping +65% FY26 revenue; even mid-teens forward growth at high margins leaves the price stretched. 25% as the plausible 'grows but normalizes' path.		Requires hyperscaler + sovereign capex to keep compounding and the cadence to ship without a share break - consistent with a sold-out backlog and FQ2 guide ~\$91B, but not a step-change beyond it. 32% as the central, still-negative outcome at a \$5T cap.		Demand outrunning supply, +92% Data Center, +199% networking and a sold-out Blackwell Ultra backlog make sustained compounding credible if the capex cycle doesn't pause. 20% weight on capex holding and Rubin extending the run.		Everything works - capex compounds for years AND the cadence holds AND CUDA/networking entrench the platform AND ASICs/AMD stay marginal. ~8%, the long-duration AI-infrastructure tail.	
WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS	
The AI capex cycle hits a digestion air-pocket: hyperscalers pause after over-building, Data Center revenue contracts (~10% then -5%), and custom ASICs plus AMD's H2-2026 launches take share at the margin so gross margin compresses toward the low-70s. China stays a structural \$0. The compute giant cyclically resets to a slower-growth, lower-margin base.		Demand keeps growing but decelerates fast as the first wave of build-out matures: +12% then high-single-digit as custom ASICs and AMD chip away GPU share and pricing, holding gross margin in the high-50s/low-60s operating. The annual cadence ships but each generation no longer doubles the installed base; sovereign + diversified demand cushions, not offsets.		The modal path: datacenter AI capex holds, Blackwell Ultra ramps into Rubin on the annual cadence, and Data Center compounds (+40% then decelerating) at a defended ~60% operating margin while networking (+199% last quarter) extends the full-rack systems franchise. Revenue roughly doubles over the window; China remains \$0. A genuinely dominant, fast-growing compute platform - but already priced for it.		Hyperscaler capex holds and broadens: the Rubin generation extends the annual cadence, networking and full-rack systems compound the moat beyond the GPU, sovereign deployments (Saudi HUMAIN ~\$20B, UAE) and diversified AI clouds (~50% of Data Center) add a durable second leg, and gross margin stays mid-70s. Demand keeps outrunning supply; CUDA + the developer ecosystem hold share against ASICs and AMD.		The full second act: AI infrastructure spend compounds for years across hyperscale, sovereign and enterprise, NVIDIA stays the platform standard through Rubin and beyond, networking + software + the CUDA ecosystem entrench an end-to-end systems moat, and gross margin holds low-to-mid-70s as supply scales. Revenue more than doubles at peak margins with no meaningful share loss.	
A cyclical semiconductor whose growth has rolled over earns a mid-teens terminal multiple → DCF ~\$40 (-80%). Net cash is immaterial against a ~\$5T cap; this is what a priced-for-perfection name does when perfection breaks.		DCF ~\$77 (-63%) - a still-large, still-profitable franchise whose hyper-growth has normalized is worth far below a \$5T price; the market's premium assumes years of compounding the bear doesn't deliver.		DCF ~\$159 (-23%) - even with capex holding and the cadence intact, ~50x FCF and a ~19x terminal multiple on a company this size leaves the modal case below today's price; the franchise is excellent and the entry is rich.		DCF ~\$246 (+20%) - if capex holds and the cadence extends, the franchise grows into and past the multiple; this is the core of the bull and where the price is defensible.		DCF ~\$355 (+73%) - the build-out runs and NVIDIA captures it as the standard platform; ~8% probability, the asymmetric upside the price embeds beyond the modal case.	

NVIDIA business snapshot

Bear \$76.59 Base \$158.65 Bull \$245.72

FY22-FY26 history + FY27-FY31 scenario projections · fiscal years end late January · FQ1 FY2027 (rep. May 2026) results



Sources: SEC XBRL company facts (CIK 1045810; revenue, GAAP operating income/margin, cash flow, debt), NVIDIA FQ1 FY2027 release (revenue \$81.6B +85% YoY, Data Center \$75.2B +92% with compute \$60.4B / networking \$14.8B +199%, non-GAAP gross margin 75.0%, FQ2 guide ~\$91B, zero China datacenter revenue), Blackwell Ultra (GB300) sold out through mid-2026. Revenue modeled as a growth path off FY26; FCF = revenue x FCF margin; Gordon terminal at scenario WACC. EV/sales & EV/FCF vs peers.

NVIDIA 7 Powers · the moat, scored

Bear \$76.59 Base \$158.65 Bull \$245.72

Arena. AI accelerators / datacenter compute - NVIDIA (GPUs + networking + the CUDA software platform, ~88% Data Center) vs AMD, the hyperscaler custom-ASIC programs (OpenAI-Broadcom, in-house silicon), and Broadcom; auditing a cornered-resource (CUDA) + scale Power against custom silicon and a credible #2.

POWER AUDIT

dominant-power durability: high

POWER AUDIT · 7 POWERS

Scale Economies ■■■	Massive R&D and supply scale at \$215.9B revenue funds a once-a-year architecture cadence rivals struggle to match; the leader's volume self-reinforces.
Network Economies ■■■	The CUDA developer ecosystem - millions of developers, libraries, frameworks tuned to NVIDIA - is a genuine two-sided network that compounds with installed base.
Counter-Positioning ■ ■ ■	None meaningful now - NVIDIA is the incumbent being attacked by lower-cost ASICs/AMD, not the counter-positioner.
Switching Costs ■■■	Code, models and full racks are written/optimized to CUDA + NVIDIA networking; re-platforming to ASICs or AMD is costly and risky - a real lock-in.
Branding ■■■	Strong reputation as the AI-compute standard, but the buy is governed by performance/TCO and ecosystem, not brand premium.
Cornered Resource · thesis leans here ■■■	The dominant Power: CUDA + the integrated GPU/networking/software stack is a ~two-decade head start rivals cannot quickly replicate - the moat's foundation.
Process Power ■■■	Deep systems/packaging and supply-chain (CoWoS) execution feeding the annual cadence; real, but partly bought capacity others can also access.

PEER SET

AMD The credible #2; MI450/Helios ships H2-2026 with OpenAI (6GW) and Meta (~\$100B) commitments - the direct GPU challenger.	30% gr · 25% mgn · ~30x EV/FCF
Custom ASICs (hyperscalers) In-house AI silicon (Google TPU, plus new programs) fragmenting GPU procurement; reduces NVIDIA's wallet share at the largest buyers.	· captive
OpenAI-Broadcom 10GW of custom accelerators (Broadcom-designed) landing H2-2026 - the marquee defection from merchant GPUs to bespoke silicon.	· ~partnership
Huawei (Ascend) Entrenching in China as the closed-market alternative now that NVIDIA datacenter revenue there is \$0; not a US/global threat yet.	· private

THREAT VECTORS · FALSIFIERS

Cornered Resource · Custom ASICs (OpenAI-Broadcom, hyperscaler in-house silicon) falsifier: Custom-silicon deployments take visible Data Center share and slow NVIDIA's growth/pricing as buyers re-platform off CUDA.
Switching Costs · AMD (MI450/Helios, H2-2026) falsifier: Major accounts (OpenAI, Meta) shift meaningful training/inference to AMD, proving the CUDA lock-in is breachable at scale.
Scale Economies · A datacenter AI capex digestion/air-pocket falsifier: Hyperscaler capex pauses, the annual cadence stops compounding the installed base, and the scale advantage stops self-reinforcing.

COMPETITIVE READ

NVIDIA's dominant Power is a cornered resource - CUDA plus the integrated GPU/networking/software stack, reinforced by switching costs, scale and a developer-ecosystem network - and it's durable (high). The live threat is the H2-2026 wave: AMD's MI450 and the OpenAI-Broadcom custom ASICs erode GPU share at the margin even if NVIDIA stays the standard, and China datacenter is a permanent \$0. The Audit squares with the DCF: the Power is real, but the swing factor is whether datacenter AI capex holds or hits an air-pocket, and at ~50x FCF on ~\$5T even the modal capex-holds case lands ~-23% - the distribution is the finding, weighted -25%.

NVIDIA show your work

Bear \$76.59 Base \$158.65 Bull \$245.72 · Weighted \$153.54 (-22.3%)

ULTRA BEAR\$40.37					BEAR\$76.59					BASE\$158.65					BULL\$245.72					ULTRA BULL\$355.27				
ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS				
5y revenue CAGR (%)					-0.6					5y revenue CAGR (%)					+7.6					5y revenue CAGR (%)				
Year-5 op margin (%)					51.0					Year-5 op margin (%)					58.0					Year-5 op margin (%)				
WACC (%)					10.5					WACC (%)					9.5					WACC (%)				
Terminal growth (%)					2.5					Terminal growth (%)					3.0					Terminal growth (%)				
Terminal FCF (\$B)					\$88.04B					Terminal FCF (\$B)					\$139.97B					Terminal FCF (\$B)				
Starting revenue (\$B)					215.90					Starting revenue (\$B)					215.90					Starting revenue (\$B)				
Scenario probability (%)					15					Scenario probability (%)					25					Scenario probability (%)				
5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B				
FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF
FY26	194.31	+48%	+77.72	+70.34	FY26	241.81	+55%	+101.56	+92.75	FY26	302.26	+58%	+132.99	+122.01	FY26	323.85	+60%	+145.73	+134.32	FY26	341.12	+61%	+156.92	+145.02
FY27	184.59	+48%	+73.84	+60.47	FY27	261.15	+56%	+112.30	+93.66	FY27	377.82	+59%	+170.02	+143.10	FY27	427.48	+61%	+200.92	+170.67	FY27	477.57	+62%	+229.23	+195.81
FY28	190.13	+50%	+77.95	+57.78	FY28	276.82	+57%	+121.80	+92.77	FY28	438.28	+60%	+201.61	+155.68	FY28	517.25	+62%	+248.28	+194.38	FY28	611.29	+63%	+305.65	+241.29
FY29	199.64	+51%	+83.85	+56.24	FY29	293.43	+57%	+129.11	+89.81	FY29	490.87	+60%	+230.71	+163.44	FY29	594.84	+62%	+291.47	+210.32	FY29	739.66	+63%	+377.23	+275.23
FY30	209.62	+51%	+88.04	+53.44	FY30	311.04	+58%	+139.97	+88.91	FY30	539.96	+60%	+253.78	+164.94	FY30	666.22	+62%	+326.45	+217.10	FY30	865.40	+63%	+441.36	+297.61
Σ PV explicit FCF					+298.27					Σ PV explicit FCF					+926.79					Σ PV explicit FCF				
+ PV terminal (g 2.5%)					684.71					+ PV terminal (g 3.0%)					5017.50					+ PV terminal (g 4.0%)				
EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE				
Operating EV					\$982.98B					Operating EV					\$5944.29B					Operating EV				
+ Cash & investments					\$10.60B					+ Cash & investments					\$10.60B					+ Cash & investments				
– Total debt					\$8.50B					– Total debt					\$8.50B					– Total debt				
= Equity value					\$985.08B					= Equity value					\$5946.39B					= Equity value				
FY30 shares (M)					24,400					FY30 shares (M)					24,200					FY30 shares (M)				
= Expected per share					\$40.37					= Expected per share					\$245.72					= Expected per share				
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT				
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y	
\$66.51	\$109.57	\$180.51	\$297.37		\$120.57	\$189.81	\$298.80	\$470.39		\$244.10	\$375.58	\$577.88	\$889.14		\$369.48	\$555.57	\$835.38	\$1,256.13		\$526.86	\$781.32	\$1,158.69	\$1,718.32	
0.34×	0.55×	0.91×	1.51×		0.61×	0.96×	1.51×	2.38×		1.24×	1.90×	2.92×	4.50×		1.87×	2.81×	4.23×	6.36×		2.67×	3.95×	5.86×	8.70×	



NVIDIA

People · Opportunity · Context · Deal

Bear \$76.59 Base \$158.65 Bull \$245.72

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Jensen Huang · founder-led 33 yrs in seat

3.77% HIGH
insider ownership key-person risk

Capital allocation (realized)
Buyback-led: a \$50.0B authorization (Aug 2024) then an additional \$60.0B (Aug 2025), both no-expiration; token quarterly dividend of \$0.01/share (post-10:1 split); R&D-intensive (FY2025 R&D up sharply on more compute/infra and new-product engineering); SBC \$4.74B in FY2025 (+33% YoY); M&A record mixed — Mellanox (\$6.9B, closed 2020) completed, the \$40B Arm acquisition abandoned 2022 on antitrust opposition.

Incentive alignment
FY2025 total comp \$49.9M: base salary \$1.5M (first raise in 10 years), non-equity incentive \$6.0M, stock awards \$38.8M, ~\$3.5M other (security, driver, vehicle). Equity granted as RSUs plus PSUs tied to corporate financial-performance targets and market-condition (TSR) PSUs.

GOVERNANCE FLAGS

- single class of common stock, one vote per share; no preferred outstanding; no cumulative voting
- Huang ~3.77% economic stake (~923M of ~24.46B shares, Mar 2025) — control rests on the operating role, not super-voting shares
- no separate independent Chair; Huang is President & CEO and the only non-independent director — mitigated by a designated Lead Independent Director running executive sessions without the CEO
- board of 13 with 12 independent directors

PEOPLE READ

Founder-CEO since 1993 (~33-year tenure) with a ~3.77% economic stake under a clean single-class one-share-one-vote structure and a 12-of-13 independent board with a Lead Independent Director; capital return is buyback-led (\$110B authorized across 2024-2025) with a token dividend, and Huang is the singular face of strategy, so key-person concentration is high.

THE FOUR LEGS

People observable composite · 1–5

Opportunity
The scenario distribution · the business snapshot · the Arthur Indicator

Context
The 7 Powers analysis (Power Audit)

Deal
Open-market common — entry price weighed against the probability-weighted fair value (the -25.2% finding), then position-sized.

NVIDIA supporting analysis · disclaimers · glossary

Bear \$76.59 Base \$158.65 Bull \$245.72

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

~50x FCF on the world's largest company.
~\$96.7B FCF and ~60% operating margins, but at a ~\$5T cap the modal DCF is still ~-23% - the price, not the business, is the constraint; the distribution does the talking.
- 2

The single swing factor is datacenter capex.
The whole valuation hinges on hyperscaler + sovereign AI spend compounding for years; capex holding is the bull (+20%), a digestion/air-pocket is the ultra-bear (~80%).
- 3

China datacenter is a structural \$0.
Results and guidance assume ~no China; Beijing steered firms away and Huawei is entrenching - a permanently lost ~\$15-50B/yr TAM already out of the numbers, not a future risk.
- 4

Competition all lands H2-2026.
AMD MI450/Helios (OpenAI 6GW, Meta ~\$100B) and OpenAI-Broadcom 10GW custom ASICs ship H2-2026; the GPU monopoly erodes at the margin even if NVIDIA stays the leader.
- 5

The systems moat is more than the GPU.
Networking +199% and CUDA + the developer ecosystem make the full-rack platform stickier than a chip - the durability case for the bull/ultra-bull, but already in the price.

FALSIFICATION TRIGGERS

Bull validation hyperscaler capex guidance keeps rising · Rubin samples Q4-2026 / volumes Q1-2027 on cadence · Data Center + networking stay strong · gross margin holds mid-70s	Bear validation a major hyperscaler cuts or pauses AI capex · custom ASICs / AMD take visible Data Center share · gross margin slips toward low-70s · backlog/booking commentary softens	Reframe needed if datacenter AI capex hits a sustained air-pocket or custom silicon structurally takes share, re-rate the terminal toward a cyclical-semiconductor multiple
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Forward-looking statements. Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.	Author may hold positions. The author may own, intend to acquire, or hold short exposure to \$NVDA or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.	Do your own research. Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Data Center — NVIDIA's dominant segment (~88% of revenue, +92% last quarter) - GPUs + networking; the thesis rides on its growth.	Hyperscaler capex — Cloud-provider (Microsoft/Amazon/Google/Meta) AI-infrastructure spending - the demand source and biggest swing.	Annual cadence (Blackwell → Rubin) — NVIDIA's once-a-year architecture rhythm (Blackwell → Ultra/GB300 → Rubin); how it stays ahead of ASICs.
Custom ASICs — Purpose-built AI chips (OpenAI-Broadcom, hyperscaler in-house) fragmenting GPU procurement; the live threat at the margin.	FCF margin — Free cash flow / revenue (~45% FY26); the cash-conversion swing factor.	